

AHA SCIENTIFIC STATEMENT

Secondary Prevention After Coronary Artery Bypass Graft Surgery: 2026 Update: A Scientific Statement From the American Heart Association

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ABSTRACT: Coronary artery bypass grafting is a well-established, durable, and safe surgical intervention. However, coronary artery disease continues to progress after the procedure. Patients who have undergone bypass surgery present unique challenges in terms of secondary prevention resulting from the often severe and diffuse nature of their coronary disease, the complexities of their postoperative recovery, the burden of their comorbid conditions, and the importance of ensuring long-term graft patency and preventing further disease progression. New evidence and advances in secondary prevention strategies in the post-coronary bypass grafting population have emerged since the American Heart Association's 2015 scientific statement on this topic. Secondary prevention strongly correlates with improved outcomes after bypass surgery, providing the rationale and urgency for this updated scientific statement to promote evidence-based practical considerations and to improve their use.

Key Words: AHA Scientific Statements ■ continuity of patient care ■ coronary artery bypass ■ coronary artery disease ■ diabetes mellitus ■ rehabilitation ■ secondary prevention

Coronary artery bypass grafting (CABG) is a well-established, durable, and safe surgical intervention. However, coronary artery disease (CAD) continues to progress after the procedure, necessitating aggressive lifelong management.¹ Patients who have undergone CABG present unique challenges in terms of secondary prevention resulting from the often severe and diffuse nature of their CAD, the intricacies of their postoperative recovery, the burden of comorbid conditions, and the importance of ensuring long-term graft patency and of preventing further disease progression and recurrent cardiovascular events (Figure 1). Even in clinical trials, adherence to secondary prevention has been suboptimal after CABG,^{2–4} and large registries suggest a strong inverse correlation between uptake rates and mortality.^{5,6} Therefore, enhancing secondary prevention strategies in post-CABG patients has a major public health impact, potentially improving outcomes for millions of individuals. New evidence and advancements in secondary prevention strategies in patients after CABG have

emerged since the American Heart Association's 2015 scientific statement on this topic,⁷ providing the rationale and urgency for this updated scientific statement aimed at promoting evidence-based practical considerations and improving their use.

ANTITHROMBOTIC THERAPY

After CABG, antithrombotic therapy aims to prevent cardiovascular events and bypass graft failure. However, dual antiplatelet therapy (DAPT) and the addition of an oral anticoagulant (OAC) to antiplatelet therapy increase bleeding risk.⁸ Therefore, selecting the appropriate antithrombotic strategy requires careful weighing of the anticipated anti-ischemic benefits against bleeding risks.

Aspirin

Aspirin after CABG has been associated with a reduced risk of death and ischemic complications^{8,9} and



Figure 1. Main areas of secondary prevention after coronary artery bypass graft surgery.

LV indicates left ventricular; and MI, myocardial infarction.

reduced graft occlusion.¹⁰ A low (100 mg) to medium (325 mg) dose of aspirin should be initiated within 6 hours after CABG and continued indefinitely to improve graft patency and reduce future cardiovascular events (Class 1; Level of Evidence A).^{11–13} Low-dose aspirin appears sufficient because it exceeds the minimal effective dose required for platelet thromboxane A_2 suppression,¹⁴ whereas higher aspirin doses increase the risk of gastrointestinal bleeding.^{15,16} Although continuing aspirin perioperatively has been associated with higher perioperative blood loss and transfusions (but not with an increased risk of surgical re-exploration),⁹

patients already taking aspirin may continue taking it until surgery to reduce the risk of perioperative myocardial infarction (MI).¹⁸ However, there is no evidence of benefit with immediate preoperative (<24 hours) initiation of aspirin in patients undergoing elective CABG who are aspirin naïve or in whom aspirin was discontinued.^{13,18,19}

In patients with aspirin hypersensitivity, aspirin desensitization can be considered. If unfeasible or unsuccessful, clopidogrel monotherapy may be entertained, although evidence in patients undergoing CABG is limited.²⁰

P2Y₁₂ Inhibitors

Post hoc data from randomized controlled trials (RCTs) in acute coronary syndrome (ACS) patients after CABG indicate that DAPT improves anti-ischemic efficacy over aspirin alone, supporting its use for up to 1 year postoperatively in this high-risk subgroup.^{21–23} In the CURE trial (Clopidogrel in Unstable Angina to Prevent Recurrent Ischemic Events), DAPT with clopidogrel resulted in a 1.7% absolute reduction in major adverse cardiovascular events (MACEs) compared with aspirin.²¹ The PLATO trial (Platelet Inhibition and Patient Outcomes) showed that ticagrelor reduced all-cause and cardiovascular mortality rates by 50% at 1 year after CABG compared with clopidogrel.²² The TACSI trial (Dual Antiplatelet Therapy With Ticagrelor and Acetylsalicylic Acid vs. ASA Only After Isolated CABG in Patients With Acute Coronary Syndrome), a registry-based RCT including 2201 patients with ACS undergoing CABG, found no difference in the composite of death, MI, stroke, or repeat revascularization at 1 year between DAPT with ticagrelor and aspirin.²³ However, this trial was limited by high treatment discontinuation rates.

There is no definitive evidence that DAPT provides additional clinical benefits over aspirin in post-CABG patients with stable, chronic CAD. However, DAPT in these patients may lower graft occlusion rates.^{13,24,25} A patient-level meta-analysis combining RCT data from 871 patients (1064 with saphenous vein graft) showed that 1 year of DAPT with ticagrelor was associated with a nearly 50% reduction of saphenous vein graft failure compared with aspirin (11.2% versus 20%; odds ratio, 0.51 [95% CI, 0.35–0.74]; $P < 0.001$).²⁶ However, the risk of clinically important bleeding was significantly increased with DAPT, and no benefit was observed in terms of ischemic events. The effect of DAPT on early graft failure appears consistent for both arterial and vein grafts.²⁴ The DACAB-FE study (Different Antiplatelet Therapy Strategy After Coronary Artery Bypass Grafting Follow-Up Extension) found a lower risk of MACEs at 5 years in patients who received DAPT with ticagrelor during the first year after CABG.²⁷ Some limited evidence supports the use of DAPT to improve saphenous vein graft patency after off-pump CABG²⁶ and in high-complexity CAD.²⁸

Ticagrelor monotherapy after CABG did not reduce the incidence of clinical events or graft failure compared with aspirin, although dedicated RCTs were underpowered.^{29–31}

Anticoagulants

Among patients undergoing CABG in the COMPASS trial (Cardiovascular Outcomes for People Using Anticoagulation Strategies),³² rivaroxaban combined with low-dose aspirin was more effective than aspirin alone for the prevention of MACEs but was not associated with

a reduction in graft failure, likely reflecting the platelet-driven mechanism of early graft failure.

Postoperative Atrial Fibrillation

Given the high stroke risk in patients with postoperative atrial fibrillation (AF), initiation of OAC requires evaluation of the individual patient risk-to-benefit ratio.³³ Observational studies have shown that early postoperative OAC initiation was not associated with reduced risks of ischemic stroke or thromboembolism but with increased risk of major bleeding and mortality.^{34,35} A meta-analysis of 28 observational studies involving 1 698 307 patients undergoing CABG found no significant difference in thromboembolic risk (effect size on hazard ratio, -0.11 ; -0.38 to 0.13) but higher bleeding risk (effect size on hazard ratio, 0.32 ; 0.06 – 0.58) in patients with compared with those without OAC.³⁶

Practical Considerations for Antithrombotic Therapy After CABG

- Low-dose aspirin initiated within 6 hours after CABG and continued indefinitely improves graft patency and reduces future cardiovascular events; higher doses offer no incremental benefit and increase gastrointestinal bleeding risk.
- In patients with ACS undergoing CABG, DAPT for 1 year may be associated with a lower risk of death and cardiac events.
- In patients with chronic coronary disease, routine DAPT is not indicated. DAPT with aspirin and clopidogrel or ticagrelor can be considered for 1 year to prevent graft failure in patients who are not at high bleeding risk.

LIPID MANAGEMENT

Atherosclerosis progression in bypassed or native coronary arteries is exacerbated by continued smoking, suboptimal adherence to aspirin therapy, and abnormal lipid profiles (high low-density lipoprotein cholesterol [LDL-C], low high-density lipoprotein cholesterol, or high triglyceride levels).^{37,38} Aggressive LDL-C lowering has been shown to reduce graft atherosclerosis progression and repeat revascularization rates.³⁸ Many post-CABG patients do not attain lipid targets despite high-intensity statins,^{39,40} and LDL-C levels at 1-year after CABG have been independently associated with long-term MACEs.⁴¹ Patients with prior CABG who develop an ACS have a markedly elevated risk of recurrent MACEs and death ($\approx 28\%$ and $\approx 9\%$, respectively, at a median of 2.8 years) even on maximal statin therapy.⁴² Although high-intensity statin therapy is prescribed for the majority of patients, the addition of nonstatin LDL-C-lowering therapies is often required to attain contemporary guideline-directed lipid targets. According to the overall evidence in post-CABG

patients, an LDL-C of ≤ 70 mg/dL (≤ 1.8 mmol/L), a non-high-density lipoprotein of ≤ 93 mg/dL (≤ 2.4 mmol/L), and an apolipoprotein B of ≤ 0.70 g/L represent minimum targets to achieve.⁴³

The IMPROVE-IT trial (Improved Reduction of Outcomes: Vytorin Efficacy International Trial) demonstrated an 8.8% absolute risk reduction in MACEs with simvastatin/ezetimibe compared with simvastatin alone in post-CABG patients, with a number needed to treat of 11.³⁹ PCSK9 (proprotein convertase subtilisin/kexin type 9) inhibitors further enhance lipid control in high-risk patients. In the ODYSSEY OUTCOMES trial (Evaluation of Cardiovascular Outcomes After an Acute Coronary Syndrome During Treatment With Alirocumab), patients undergoing CABG and treated with alirocumab achieved LDL-C levels of 25 to 50 mg/dL (0.65 to 1.3 mmol/L) and had a 6.4% absolute risk reduction in MACEs compared with those given placebo (number needed to treat, 16).⁴² However, these patients continued to exhibit a residual MACE rate of 24.5% at 4 years, highlighting persistent risks despite intensive lipid lowering.

Icosapent ethyl reduces cardiovascular events in high-risk patients with hypertriglyceridemia on statin therapy. In REDUCE-IT (Reduction of Cardiovascular Events With Icosapent Ethyl—Intervention Trial), 1837 patients (22.5%) had a history of CABG; 897 were randomized to icosapent ethyl, and 940 were randomized to placebo.⁴⁴ Randomization to icosapent ethyl led to a significant 24% reduction in cardiovascular death, MI, stroke, coronary revascularization, or hospitalization for unstable angina and a significant 31% reduction in MACEs compared with placebo.⁴⁴ A substantial number of post-CABG patients ($\approx 20\%$ – 25%) are potentially eligible for icosapent ethyl therapy,⁴⁵ but the therapy remains controversial and may be considered on an individual basis.

In the selection of therapy targets, although numerous lipid management options have been studied, not all have reported cardiovascular outcomes in post-CABG patients (eg, bempedoic acid, inclisiran). In our view, the post-CABG population is at high risk of subsequent events resulting from surgical, severe, and often diffuse CAD, frequently in the presence of saphenous vein grafts. We therefore consider it reasonable to apply in post-CABG patients the 2019 European Society of Cardiology/European Atherosclerosis Society guidelines and the 2021 European Society of Cardiology guidelines on cardiovascular disease prevention in clinical practice, which suggest a target of < 55 mg/dL (< 1.4 mmol/L) for very high-risk patients.^{46,47} Somewhat more conservatively, the “2023 AHA/ACC/ACCP/ASPC/NLA/PCNA Guideline for the Management of Patients With Chronic Coronary Disease” and the “2025 AHA/ACC Clinical Performance and Quality Measures for Patients With Chronic Coronary Disease” on the management of dyslipidemia indicate that the LDL target should be a

50% reduction from baseline and levels of < 70 mg/dL (< 1.8 mmol/L).^{48,49}

Practical Considerations for Lipid Management After CABG

- Treat to an LDL-C threshold of 55 mg/dL (1.4 mmol/L) as an ideal target and to no higher than 70 mg/dL (< 1.8 mmol/L) as an established target. One may consider an alternative target of non-high-density lipoprotein of < 85 mg/dL (2.2 mmol/L) or an apolipoprotein B target of ≤ 0.65 g/L.
- First-line therapy: High-intensity or maximally tolerated statin therapy aimed at achieving LDL-C target levels.
- Second-line therapy: If the LDL-C target is not achieved with statins alone, ezetimibe.
- Third-line therapy: If the combination of statin and ezetimibe still fails to achieve the target, PCSK9 inhibitor, or consider icosapent ethyl if the triglyceride level is between 135 mg/dL (1.5 mmol/L) and 500 mg/dL (5.6 mmol/L).

β -BLOCKER THERAPY

Preoperative β -Blockers

Evidence to support the Society of Thoracic Surgeons quality measure requiring β -blocker administration within 24 hours before CABG is uncertain in β -blocker-naïve patients.^{50,51} These metrics, established on a historical cohort study demonstrating lower rates of 30-day post-CABG mortality, have since been challenged by contemporary studies demonstrating no benefit,^{51–57} especially in the context of nonemergent CABG without recent MI.⁵²

Perioperatively, β -blockers are useful for the prevention of AF after CABG (Class 1; Level of Evidence B-R),¹³ especially when initiated ≤ 72 hours postoperatively in high-risk patients (age > 75 years, history of AF, longer aortic cross-clamp duration).⁵⁸ If the patients is already treated with β -blockers before admission, perioperative continuation is associated with lower 30-day mortality only in those with reduced left ventricular ejection fraction (LVEF) $\leq 40\%$, as well as with a reduction in postoperative AF and ventricular arrhythmias.^{58–60}

Long-Term Indications

Evidence for long-term continuation of β -blockers after CABG is largely extrapolated from the medical literature^{57,61} wherein selective β -blockade reduces MACEs in those with LVEF $\leq 40\%$ with or without previous MI (Class 1; Level of Evidence A) and possibly in those with LVEF $\leq 50\%$.⁴⁸ Contemporary cohort studies showed lower rates of MACEs with cardioselective β -blockers after CABG, which is consistently driven by a reduction in

MI across subgroups with previous MI, heart failure, AF, and reduced LVEF.^{62,63} However, a large SWEDEHEART (Swedish Web-System for Enhancement and Development of Evidence-Based Care in Heart Disease Evaluated According to Recommended Therapies) registry study did not show a correlation between β -blocker use and mortality after CABG.⁶

Practical Considerations for β -Blocker Therapy After CABG

- Preoperative: No evidence supports β -blocker administration within 24 hours before elective CABG in β -blocker-naïve patients.
- Perioperative: β -blockers are used to prevent AF in high-risk patients. If already on β -blockers, perioperative continuation is associated with a reduction in AF, as well as with lower 30-day mortality and ventricular arrhythmias in those with LVEF $\leq 40\%$.
- Long term: Cardioselective β -blockers after CABG are associated with lower rates of MACEs but no reduction in mortality.

ANTIHYPERTENSIVE THERAPY

The activation of numerous neurohumoral and endogenous vasodilator systems occurs early after CABG.⁶⁴ Because the post-CABG population is at high risk for cardiovascular events, clinicians and patients should strive for the strictest blood pressure goals, that is, $<130/<80$ mmHg.^{48,49} In terms of classes of medications, the IMAGINE trial (Ischemia Management With Accupril Post-Bypass Graft via Inhibition of the Converting Enzyme) showed no clinical benefit over 3 years for the early initiation of angiotensin-converting enzyme (ACE) inhibitor therapy early after CABG and, in fact, an increase in cardiac events within the first 3 months after CABG.⁶⁵ In contrast, data from the SWEDEHEART registry after isolated first-time CABG suggested that ACE inhibitor therapy may be beneficial for all patients who can tolerate it. In this low-risk cohort, 91% ($n=28\,782$) had at least one Class I indication for ACE inhibitor/angiotensin receptor blocker (ARB) therapy (previous MI, diabetes, heart failure, reduced LVEF, and hypertension).⁶⁶ Therefore, the use of an ACE inhibitor during this period is best individualized and reassessed over time.

Radial Artery Grafting

The role of calcium channel blockers after radial artery grafting remains debated.⁶⁷ RCTs have been small, primarily used diltiazem (which may be limited by cardiac and venous effects), and showed overall no benefit.^{68,69} Conversely, an observational meta-analysis that pooled data from 6 randomized trials (732 patients with 502 on calcium channel blockers and 2 trials specifying the use of

a dihydropyridine) showed large effect sizes with a 48% reduction in MACEs and 80% reduction in radial graft occlusion evaluated by angiography at the midterm follow-up.⁷⁰ We support the use of calcium channel blockers for the first postoperative year for radial artery grafts, with dihydropyridines preferred over diltiazem or verapamil.

Practical Considerations for Antihypertensive Therapy After CABG

- ACE inhibitors or ARBs are indicated for patients who have a clinical indication other than CABG (previous MI, diabetes, heart failure, reduced LVEF, and hypertension).
- The use of calcium channel blockers for the first postoperative year, with dihydropyridines (eg, amlodipine or nifedipine) preferred over nondihydropyridines (eg, diltiazem or verapamil), may help limit radial artery graft spasm.

MANAGEMENT OF PREVIOUS MI AND LEFT VENTRICULAR DYSFUNCTION

After CABG, guideline-directed medical therapy (GDMT) in the context of previous MI and persistent reduced LVEF $<40\%$ includes renin-angiotensin-aldosterone system inhibitors as first-line therapy to reduce mortality and heart failure hospitalization, to prevent reverse remodeling, and to improve functional status.^{71–73} The use of an angiotensin receptor-neprilysin inhibitor (ARNi) is recommended as first-line therapy (Class 1; Level of Evidence B-R), with an ACE inhibitor considered next or the use of an ARB if the patients is intolerant of ARNi and ACE inhibitors (Class 1; Level of Evidence A).^{74,75} β -Blockers and ACE inhibitors may work synergistically to improve LVEF and end-diastolic diameter more than ACE inhibitors alone.^{76,77} Furthermore, in patients with New York Heart Association (NYHA) class II to III heart failure with reduced ejection fraction who tolerate an ACE inhibitor or ARB, replacement with ARNi is advised. In patients with persistent NYHA class II to IV symptoms, LVEF $<35\%$, estimated glomerular filtration rate ≥ 30 mL $\cdot$ min $^{-1}\cdot 1.73$ m $^{-2}$, and serum potassium <5.0 mEq/L, the addition of a mineralocorticoid receptor antagonist (MRA) is recommended according to the 2022 ACC/AHA guideline (Class 1; Level of Evidence A)⁷⁵ and the European Society of Cardiology guidelines (Class II; Level of Evidence B),⁷⁸ which expand the indication for MRAs to those with mildly reduced LVEF as part of the recommended intensive predischARGE and early postdischarge follow-up of patients hospitalized for acute heart failure (Class 1, Level of Evidence B).⁷⁸ Sodium-glucose cotransporter 2 (SGLT2) inhibitors reduce cardiovascular death and heart failure hospitalization in patients with symptomatic LVEF $\leq 40\%$ regardless of diabetes status (Class 1; Level of Evidence A).^{75,78}

Evidence from nonsurgical populations favors the simultaneous initiation over traditional stepwise approaches of all 4 agents (ie, ARNi or ACE inhibitor, β -blocker, MRA, and SGLT2 inhibitor) in the context of previous MI and reduced LVEF.^{79,80} The inpatient setting allows supervised titration and early detection of adverse effects, and starting GDMT in the hospital may improve adherence by patients and continuity by primary health care professionals.^{79–81} Preoperative GDMT is the strongest predictor of its continuation on discharge after CABG (odds ratio, 7.1 [95% CI, 5.0–9.9]),⁸¹ emphasizing the value of preoperative optimization. Nevertheless, the optimal timing of GDMT initiation after surgery remains unclear. Surgical teams must leverage team-based strategies such as early follow-up, telemedicine, cardiac rehabilitation (CR), and coordinated handoffs to primary health care professionals to support timely implementation.

Device Therapy

An implantable cardioverter defibrillator is indicated for the primary prevention of sudden cardiac death in patients with LVEF $\leq 40\%$ and inducible sustained ventricular tachycardia or ventricular fibrillation (Class 1; Level of Evidence B-R), as well as ≥ 40 days after MI and ≥ 90 days after revascularization in patients with LVEF $< 35\%$ with NYHA class II to III symptoms despite GDMT or LVEF $\leq 30\%$ despite GDMT if meaningful survival of > 1 year is expected (Class 1; Level of Evidence A).^{82–84} Recent studies demonstrated a potential survival benefit of earlier implantation after CABG⁸⁵ but not after percutaneous coronary intervention.^{86,87}

Practical Considerations for Previous MI and Left Ventricular Dysfunction Management After CABG

- In those with previous MI and persistent LVEF $< 40\%$, use of an ARNi is considered first line. ACE inhibitors are used if an ARNi is not feasible, and an ARB is used if the patient is ACE inhibitor intolerant, in combination with β -blockers. If persistent with NYHA class II to IV heart failure, LVEF $< 35\%$, and estimated glomerular filtration rate ≥ 30 mL $\cdot$ min $^{-1}$ $\cdot$ 1.73 m $^{-2}$, an MRA is recommended. European Society of Cardiology guidelines further recommend adding an MRA as a predischARGE medication in those hospitalized with acute heart failure. SGLT2 inhibitors are beneficial in patients with symptomatic LVEF $\leq 40\%$.
- If feasible, all 4 agents (ie, ARNi or ACE inhibitors, β -blockers, MRA, and SGLT2 inhibitors) may be simultaneously initiated. Team-based strategies such as early follow-up, telemedicine, CR, and coordinated handoffs to primary

health care professionals support safe and timely implementation.

- Device therapy: An implantable cardioverter defibrillator is indicated for primary prevention of sudden cardiac death ≥ 40 days after MI or ≥ 90 days after revascularization in patients with LVEF $< 35\%$ and with NYHA class II to III heart failure on GDMT, LVEF $\leq 30\%$ on GDMT and LVEF $\leq 40\%$ and inducible sustained ventricular tachycardia or ventricular fibrillation. After CABG, implantation earlier than 90 days postoperatively may be considered.

DIABETES

Perioperative Glycemic Management

Preoperative hemoglobin A1c (HbA1c) levels $< 6.5\%$ correlate with fewer perioperative complications.⁸⁸ Reviews of current practice suggest diabetes screening in all patients and interventions to achieve an HbA1c level $< 7\%$ when applicable, although preventing perioperative hypoglycemia is also an important consideration.^{88,89} In addition, the use of epidural analgesia during cardiac surgery has been linked to a reduced incidence of hyperglycemia.^{88,89}

SGLT2 inhibitors and glucagon-like peptide-1 receptor agonists (GLP-1 RAs) and dual glucose-dependent insulinotropic polypeptide–GLP-1 RAs have important perioperative implications. To reduce the risk of diabetic ketoacidosis, it is prudent to discontinue SGLT2 inhibitors (if clinically feasible) ≈ 3 days preoperatively and then restart them postoperatively once oral intake (including carbohydrate intake) has resumed.^{89,90} Diabetic ketoacidosis associated with SGLT2 inhibitors often presents with relative euglycemia. GLP-1–based therapies delay gastric emptying and may increase the risk of aspiration during general anesthesia.^{89,90} These agents are typically discontinued 1 week before surgery and resumed once bowel motility has been restored postoperatively.

Long-Term Secondary Prevention Therapies in Diabetes and Obesity for Cardiometabolic Risk Reduction

SGLT2 inhibitors and GLP-1 RAs are strongly recommended for individuals with type 2 diabetes and atherosclerotic cardiovascular disease (ASCVD), including those with prior CABG, on the basis of evidence from multiple clinical trials.^{91–93} In subanalyses of the EMPA-REG OUTCOME study (BI 10773 [Empagliflozin] Cardiovascular Outcome Event Trial in Type 2 Diabetes Mellitus Patients), in post-CABG patients with type 2 diabetes, the hazard ratios for the SGLT2 inhibitor empagliflozin compared with placebo were 0.52 (95% CI, 0.32–0.84) for cardiovascular mortality, 0.57 (95%

CI, 0.39–0.83) for all-cause mortality, 0.50 (95% CI, 0.32–0.77) for hospitalization for heart failure, and 0.65 (95% CI, 0.50–0.84) for incident or worsening nephropathy.⁹⁴ The GLP-1 RAs have also been shown to reduce MACEs and body weight in patients with type 2 diabetes and established ASCVD,^{91–93} but CABG-specific data are not published. In post-CABG patients with type 2 diabetes, efforts should focus on incorporating 1 of these 2 classes of medications (or a combination) as secondary prevention. In people with diabetes and concomitant peripheral artery disease, semaglutide improves maximum and pain-free walking distance, improves ankle-brachial index, and reduces disease progression as evaluated in the STRIDE trial (A Research Study to Compare a Medicine Called Semaglutide Against Placebo in People With Peripheral Arterial Disease and Type 2 Diabetes).⁹⁵

SGLT2 inhibitors, although originally developed to improve HbA1c, have been shown to reduce cardiovascular death and heart failure hospitalizations in people with heart failure regardless of the presence or absence of diabetes.^{75,78} A subanalysis of the CABG cohort from DAPA-HF (Study to Evaluate the Effect of Dapagliflozin on the Incidence of Worsening Heart Failure or Cardiovascular Death in Patients With Chronic Heart Failure) and DELIVER (Dapagliflozin Evaluation to Improve the Lives of Patients With Preserved Ejection Fraction Heart Failure)⁹⁷ showed that the SGLT2 inhibitor dapagliflozin consistently reduced the primary and key secondary end points after CABG across the spectrum of LVEF in the context of symptomatic heart failure.

The GLP-1 RA semaglutide has also been shown to reduce MACEs in patients with overweight and obesity without diabetes but with established ASCVD.⁹⁸ In a subgroup analysis of patients from the SELECT trial (Semaglutide Effects on Heart Disease and Stroke in Patients With Overweight or Obesity), patients undergoing CABG were at higher risk of cardiovascular events compared with those without CABG,⁹⁹ and semaglutide led to consistent reductions in MACE outcomes regardless of CABG history.

Practical Considerations for Diabetes Management After CABG

1. For long-term secondary prevention:
 - SGLT2 inhibitors are prioritized in patients with diabetes and a history of CABG regardless of baseline HbA1c. These therapies are to be used primarily for their effect on reducing MACEs and to afford renal protection.
 - SGLT2 inhibitors are prioritized in patients after CABG with heart failure regardless of diabetes status or ejection fraction. These therapies are used primarily for their effect on reducing

cardiovascular death and hospitalization for heart failure and cardiorenal events.

- GLP-1 RAs are prioritized in post-CABG patients with diabetes or a body mass index (BMI) >27 kg/m² without diabetes to reduce MACEs regardless of HbA1c.
 - GLP-1 RAs, specifically semaglutide, are prioritized in people with diabetes and concomitant peripheral artery disease to improve functional outcomes and to prevent disease progression.
2. For perioperative considerations:
 - SGLT2 inhibitors should be avoided in people with type 1 diabetes.
 - When clinically appropriate, SGLT2 inhibitors should be withheld 3 days before CABG and restarted once oral intake, including carbohydrate intake, has been initiated postoperatively to reduce the risk of diabetic ketoacidosis.
 - When clinically appropriate, once-weekly GLP-1 RAs should be stopped 1 week before surgery and oral GLP-1 RAs should be stopped 3 days before surgery and resumed when the patient is eating well and bowel motility is restored.

SMOKING CESSATION

Cigarette smoking is the leading preventable cause of CAD and increases the risk of complications after CABG.¹⁰⁰ Abstaining from smoking after CABG can lower 5-year mortality risk by 35% and MACEs by 18%.¹⁰¹ These data underscore the urgent need for effective smoking cessation strategies as part of comprehensive care.

Smoking induces vascular calcification and matrix metalloproteinase expression, reducing saphenous vein graft patency.^{102,103} A pooled analysis of 7 trials involving 4413 patients found a direct association between smoking and graft failure.¹⁰⁴ Smoking cessation interventions combining intensive counseling, nicotine replacement therapy, and pharmacotherapy with bupropion and varenicline are most effective.¹⁰⁵ A Cochrane review showed that electronic nicotine delivery systems such as electronic cigarettes potentially improved cessation rates compared with nicotine replacement therapy¹⁰⁶; however, their long-term cardiovascular effects are not well studied, warranting cautious interpretation. It is important to note that electronic nicotine delivery systems lack US Food and Drug Administration approval for smoking cessation; thus, traditional cessation interventions remain first line.

The benefits of quitting extend beyond the effects on vascular health and graft patency to reducing risks for cancers and chronic obstructive pulmonary disease. Emerging evidence suggests that smoking cessation efforts should begin in the perioperative phase;

smokers who quit even up to 4 weeks before CABG have outcomes comparable to those of never smokers.^{100,107,108}

The cardiac surgeon is well positioned to discuss the importance of smoking cessation before surgery. Routinely identifying smoking status in the medical record and offering preoperative cessation support are essential.¹⁰⁷ The 5 “As” framework (ask, advise, assess, assist and arrange) offers a structured approach for screening and treatment.¹⁰⁷ Longitudinal smoking cessation care benefits from ongoing management by general practitioners or tobacco treatment specialists. The recent National Institute for Care and Excellence guideline for smoking cessation in pregnant women recommends using carbon monoxide monitors to assess and treat tobacco dependency and may be applicable in patients undergoing CABG.^{108,109}

Practical Considerations for Smoking Cessation After CABG

- Smoking induces graft failure and MACEs. Abstaining from smoking after CABG can lower 5-year mortality risk by 35% and MACEs by 18%.
- The cardiac surgeon should document smoking status, actively counsel on the importance of smoking cessation, and initiate smoking cessation support preoperatively. Smokers who quit even 4 weeks before CABG have perioperative outcomes comparable to those of never smokers.
- As with other chronic diseases, longitudinal smoking cessation care benefits from ongoing management by general practitioners or tobacco treatment specialists.

CR AND EXERCISE CONSIDERATIONS

CR for patients who are undergoing CABG is most robust when it begins preoperatively, continues during hospitalization, and extends into outpatient care.^{110,111} Preoperative optimization, particularly in patients with comorbidities or frailty, reduces surgical risks, whereas early postoperative CR focuses on enhanced recovery. All patients with CAD and appropriate indications (after recent MI, percutaneous coronary intervention, or CABG) should be referred to a CR program to improve outcomes (Class 1, Level of Evidence A).⁴⁸

CR is a cost-effective intervention that improves exercise tolerance, pulmonary oxygen uptake, quality of life, and psychological well-being while reducing hospitalizations, MACEs, and mortality.^{111,112} In addition, CR may mitigate complications such as postoperative cognitive impairment, observed in ≈40 % of patients undergoing CABG long term.¹¹³ Early mobilization based on deep breathing exercises and chest physiotherapy is

effective in reducing postoperative cognitive impairment and pain.¹¹⁴ Enhanced recovery after surgery protocols further optimize perioperative care, reducing postoperative morbidity (bronchopneumonia, acute respiratory distress syndrome, postoperative delirium, moderate to severe acute kidney injury), hospital stays, readmission risks, and costs.¹¹⁵ CR protocols involve inspiratory muscle training, breathing, and aerobic exercise coupled with either upper- or lower-limb exercise. For in-patients, CR administered as a combination of multiple exercise therapy showed significant improvement in 6-minute walking distance.¹¹⁶

After surgery, median sternotomy incision may be associated with sternal instability due to delayed sternal healing resulting from devascularization. An exercise program consisting of integrated abdominal contraction and upper-limb exercises is beneficial for decreasing sternal complications (reduction in sternal separation from the supine and long sitting positions) after CABG.¹¹⁷ Outpatient CR after CABG incorporates high-intensity inspiratory muscle training, resistance training, and aerobic exercise, which significantly improve functional capacity, 6-minute walking distance, and peak $\dot{V}O_2$.^{117,118} Tailored exercise prescriptions, including high-intensity interval training and moderate-intensity continuous training, enhance recovery and health outcomes by addressing patient preferences and motivations.

Telerehabilitation offers a flexible and accessible alternative for delivering CR, particularly for older adults or those unable to attend center-based programs. Telerehabilitation leverages technology to improve self-management and long-term outcomes, making it a viable addition to routine care. Home-based transitional cardiac telerehabilitation provides personalized, convenient rehabilitation services that align with patient needs.¹¹⁹ It is important to note that telerehabilitation has been shown to be noninferior to center-based CR in some randomized trials.^{120,121}

Practical Considerations for CR and Exercise After CABG

- All patients with chronic CAD and appropriate indications (after recent MI, percutaneous coronary intervention, or CABG) should be referred to CR programs within 4 weeks of discharge to improve outcomes.
- CR for patients who are undergoing CABG is most robust when it begins preoperatively, continues during hospitalization, and extends into outpatient care.
- Telerehabilitation offers a flexible and accessible alternative for delivering CR, particularly for older adults or those unable to attend center-based programs.

OTHER CARDIOVASCULAR RISK REDUCTION IN PATIENTS UNDERGOING CABG, INCLUDING OBESITY, METABOLIC SYNDROME, NUTRITION, VITAMINS, SUPPLEMENTS, AND VACCINATION

The recognition of clinical obesity as a formal, chronic disease by the recent *Lancet* commission marks a significant paradigm shift.¹²² This acknowledgment underscores the importance of understanding clinical obesity as a long-term illness that requires ongoing, multidisciplinary management to mitigate associated adverse health outcomes. Historically, obesity diagnosis relied on BMI; recently, the modern concept of “clinical obesity” has evolved, redefining this as a chronic disease state rather than a risk factor for cardiometabolic health.¹²³ This updated definition and diagnostic criteria emphasize the impact of excess adiposity on reduced organ function and limitations on activities of daily living.^{123,124} Obesity is associated with insulin resistance, hypertension, and a generalized inflammatory state. In a large cohort of patients who underwent isolated CABG, waist circumference was significantly associated with clinical adverse events independently of BMI.¹²⁵

Many patients who enroll in CR programs are overweight.¹²⁶ As CR programs increasingly focus on comprehensive secondary prevention of CAD, weight management emerges as a priority, and care must be provided without stigma.¹²⁷ The primary goals of weight management include preventing further weight gain, achieving weight reduction, and sustaining long-term weight loss. Effective interventions may target individualized exercise prescriptions that maximize caloric expenditure and dietary counseling to reduce energy intake.^{127–129} Participation in CR alone is associated with minimal weight loss.¹²⁹ In addition to diet and exercise, behavioral weight loss programs emphasize the importance of health behavior change incorporating self-efficacy, goal setting, stimulus control, and stress management.^{130,131}

The Mediterranean and DASH (Dietary Approaches to Stop Hypertension) diets are associated with long-term health benefits for cardiac patients,^{131,132} and the 2023 AHA/ACC guideline for managing patients with chronic coronary disease recommends a diet emphasizing vegetables, fruits, legumes, nuts, whole grains, and lean protein to reduce the risk of ASCVD events.⁴⁸ Plant-based diets have also been shown to be effective in lowering cardiovascular disease risk factors.^{132,133} Nutritional optimization and counseling in all phases of the care of patients undergoing CABG—preoperative, intraoperative and postoperative—can enhance surgical outcomes and support secondary prevention of ASCVD, serving as a key component of both prehabilitation and CR programs.^{134,135}

Patients with a history of CABG with obesity, including those without diabetes, are at heightened risk of cardiovascular events. In many countries, pharmacotherapy is now recommended for patients with a BMI ≥ 30 kg/m² and those with a BMI >27 kg/m² with weight-related comorbidities who have failed to achieve adequate success with lifestyle interventions.^{136–138} GLP-1 RAs (liraglutide, semaglutide, and tirzepatide), orlistat, phentermine/topiramate, and naltrexone/bupropion are currently approved for long-term weight management. Whereas the GLP-1 RAs approved for weight loss have been shown to reduce MACEs in patients with type 2 diabetes in RCTs,¹³⁹ semaglutide has also been shown to reduce MACEs in patients without type 2 diabetes.¹⁴⁰ To date, only 1 study has specifically evaluated the cardiovascular outcomes of weight loss drugs in post-CABG patients.⁹⁹ In this analysis, semaglutide compared with placebo resulted in significant and consistent reductions in MACEs, with a numerically greater absolute risk reduction (2.3% versus 1.0%) observed at 3 years in patients who have undergone CABG compared with those who have not.⁹⁹ Orlistat has not demonstrated MACE reduction in randomized clinical trials. Phentermine is contraindicated for individuals with ASCVD; the effect of naltrexone/bupropion on ASCVD is unknown. When weight loss targets cannot be achieved by lifestyle modification, pharmacotherapy could be considered and may be appropriately addressed through CR after CABG.

Large-scale studies have not demonstrated cardiovascular benefits from multivitamins or supplements.^{142–144}

The US Centers for Disease Control and Prevention recommends influenza and COVID-19 vaccinations for all adults. The influenza vaccine has been shown to lower composite cardiovascular events for patients with ASCVD.^{145,146} The Centers for Disease Control and Prevention recommends the pneumococcal vaccine for patients with ASCVD.^{147,148} There is no evidence that preoperative COVID-19 vaccination poses risks in patients with high cardiovascular risk undergoing CABG,¹⁴⁹ and post-CABG vaccination (usually at or beyond 1 month postoperatively) is considered safe with low rates of complications.¹⁵⁰ Vaccinations for all adults, including timing, should be discussed and tailored to individual risk profiles.

Practical Considerations for Other Cardiovascular Risk Reduction After CABG

- In addition to diet and exercise, behavioral weight loss programs emphasize the importance of health behavior change incorporating self-efficacy, goal setting, stimulus control, and stress management and may be incorporated when obesity is present.
- Effective strategies may target a net negative energy balance through exercise prescriptions that

maximize caloric expenditure and dietary counseling to reduce energy intake.

- Evidence supports the role of pharmacotherapy in achieving meaningful weight loss and reducing MACEs in both patients with and those without diabetes.
- Influenza and pneumococcal vaccines should be considered in individuals with ASCVD.
- Vaccinations for all adults should be discussed and tailored to individual risk profiles.

MENTAL HEALTH AND COGNITIVE IMPAIRMENT

Psychosocial risk factors such as depression, anxiety, anger/hostility, social isolation, low socioeconomic status, and chronic life stress contribute to negative health behaviors such as smoking, poor dietary choices, sedentary lifestyle, and reduced adherence to treatments.^{151–155} Among patients undergoing CABG, depression is prevalent in 19% to 37% preoperatively and often persists after surgery.¹⁵⁶ Depression and anxiety are common in CR participants, with depression strongly correlated with psychological stress and decreased adherence to secondary prevention strategies.^{152,154,155,157} Therefore, it is essential to assess psychosocial risk factors and mood disturbance through clinical interviews or with standardized tools before and after surgery.^{152,156,158} Targeted interventions, including pharmacotherapy and psychotherapy, may be implemented as per established standards.¹³

Selective serotonin reuptake inhibitors are commonly used antidepressants. They affect platelet function and are associated with a modest increase in upper gastrointestinal bleeding risk.¹⁵⁹ In patients with CAD on antiplatelet therapy, consideration should be given to concomitant management with gastroprotective agents such as proton pump inhibitors when a selective serotonin reuptake inhibitor is initiated.

Postoperative delirium and cognitive dysfunction frequently occur after CABG, adversely affecting quality of life and increasing mortality risk. Patients with preoperative depression have twice the risk of delirium, whereas those with preoperative cognitive impairment have 4-fold the risk.¹¹³ A meta-analysis identified depression, diabetes, and hypertension as additional correlates for these complications.¹¹³ Intensive management through pharmacology, education, and health behavior modification and cognitive retraining present preoperative opportunities to reduce these complications. Recent innovations of off-pump and no-touch surgical techniques minimize aortic manipulation, potentially reducing the risk of perioperative neurological injury, although robust evidence is still needed.¹⁶¹ Cardiac prehabilitation and CR programs enhance physical health and alleviate symptoms of depression, anxiety, and hostility.¹⁵⁶ Preoperative physical therapy has been shown to

reduce perioperative brain injury, suggesting an opportunity to mitigate cognitive impairments associated with CABG.¹⁶² Systematic screening for preexisting conditions can identify frailty and inform personalized rehabilitation strategies, improving outcomes and adherence to secondary prevention after CABG.^{163–167}

Practical Considerations for Mental Health Management After CABG

- Depression and anxiety are common before and after CABG and correlate with an increased incidence of delirium and other complications. It is essential to assess psychosocial risk factors and mood disturbance through clinical interviews or with standardized tools before and after surgery.
- Targeted interventions to address depression and anxiety, including pharmacotherapy and psychotherapy, may be implemented per established standards. When a selective serotonin reuptake inhibitor is initiated in patients with CAD on antiplatelet therapy, gastroprotective agents are considered.
- Cardiac prehabilitation and CR programs enhance physical health and alleviate symptoms of depression, anxiety, and hostility.
- Systematic screening for preexisting conditions can identify frailty and inform personalized rehabilitation strategies, improving outcomes and adherence to secondary prevention after CABG.

KNOWLEDGE GAPS AND FUTURE DIRECTIONS

The present scientific statement provides a strong rationale for addressing the many opportunities for optimizing therapy, behavior, and prognosis after CABG as an integral part of the lifelong management of CAD. Figure 2 summarizes the proposed interventions according to the phenotype of a patient with CABG.

The above notwithstanding, many aspects of secondary prevention after CABG still require more evidence (Table). For example, prior to the TACSI trial, the available randomized data pertaining to the use of DAPT after CABG in patients with ACS were derived with CABG as a postrandomization event.^{22,23,169,170} Therefore, the suitability to undergo CABG for those on DAPT could have resulted in selecting patients more likely to benefit (or less likely to experience harm) from DAPT. Postoperative AF remains a vexing problem for which anticoagulation strategies may not provide benefit, and it is unclear whether left atrial appendage obliteration improves early outcomes or obviates the need for OAC.¹⁷¹ The role of β -blockers remains controversial and has not been studied in adequately powered subgroups of interest after CABG such as patients with diastolic dysfunction,

Secondary Prevention After CABG

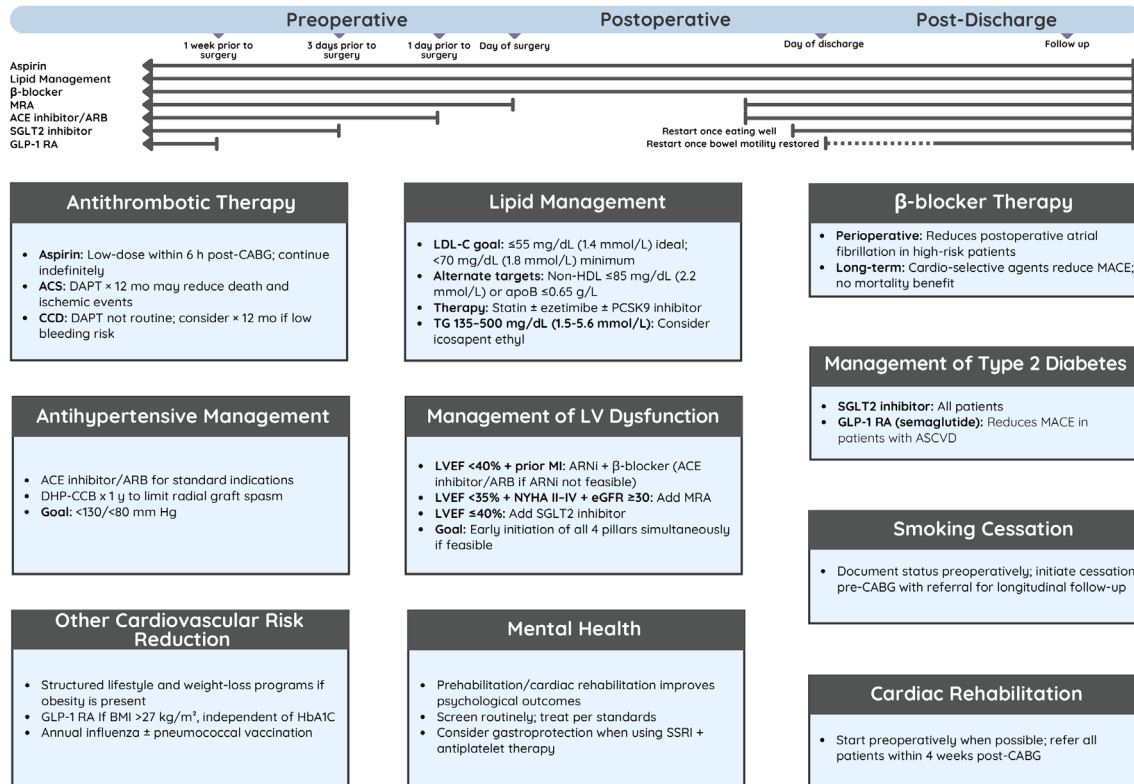


Figure 2. Disease-based opportunities and timing of secondary prevention after coronary artery bypass graft surgery.

ACE indicates angiotensin-converting enzyme; ACS, acute coronary syndrome; AF, atrial fibrillation; apoB, apolipoprotein B; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor-neprilysin inhibitor; ASCVD, atherosclerotic cardiovascular disease; BMI, body mass index; CABG, coronary artery bypass grafting; CCD, chronic coronary disease; DAPT, dual antiplatelet therapy; DHP-CCB, dihydropyridine-calcium channel blocker; eGFR, estimated glomerular filtration rate; GLP-1R, glucagon-like peptide-1 receptor agonist; HbA1C, hemoglobin A1C; HDL, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; LV, left ventricular; LVEF, left ventricular ejection fraction; MACE, major adverse cardiac events; MI, myocardial infarction; MRA, mineralocorticoid receptor antagonist; NYHA, New York Heart Association; PCSK9, proprotein convertase subtilisin/kexin type 9; SGLT2, sodium glucose co-transporter 2; SSRI, selective serotonin reuptake inhibitors; and TG, triglycerides.

postoperative AF, or complete versus incomplete revascularization. The optimal timing of implantable cardioverter defibrillator insertion after CABG is incompletely defined, with malignant events occasionally being observed early despite the clinically ubiquitous recommendation to wait 3 months, with limited data on whether myocardial viability affects the timing and decision to use an implantable cardioverter defibrillator.^{172,173} The role of optimized secondary prevention in potentiating the effects of multiarterial grafting remains unknown; relatedly, patients who cannot take antiplatelet drugs might benefit from all-arterial grafting, but this remains unproven. The prevention of delayed radial artery spasm is insufficiently studied with several unanswered questions, including, for example, the role of pharmacological agents in the presence of competitive coronary flow. The potential role of GLP-1 RAs in venous and arterial graft patency remains unknown. As a final example of knowl-

edge gaps, the overall long-term health and quality-of-life effects of polypharmacy in post-CABG patients, who typically are prescribed multiple classes of drugs, remain insufficiently elucidated.^{174,175}

Thus, future research may include patient partners to provide patient-centered perspectives, particularly emphasizing shared decision-making when treatment decisions (eg, extended DAPT or an implantable cardioverter defibrillator) involve significant trade-offs between benefit and risk.

CONCLUSIONS

Since the previous American Heart Association scientific statement on secondary prevention after CABG was published in 2015, the field has expanded tremendously, with new studies, data, and even entirely novel classes of medications. Overall, secondary prevention after CABG

Table. Knowledge Gaps in Secondary Prevention After CABG

Topic	Unanswered questions
DAPT	Benefit on graft patency Application to multi- or total arterial grafting Optimal duration and dosing
OAC	Timing/indication/duration in postoperative AF Role after successful AF ablation and left atrial appendage obliteration
Lipid management	Minimum target thresholds Role of hypertriglyceridemia treatment Inclusion of Lp(a) assessment
β-Blockers and antihypertensive treatment	Role if previous MI Calcium channel blockade in radial artery grafting How long a wait before implanting a defibrillator in severe left ventricular dysfunction after CABG Defibrillator implantation with or without myocardial viability after CABG
Diabetes management	Role of newer agents and obesity management on long-term outcomes and graft patency
Smoking cessation	Role of electronic cigarettes and cannabis products on long-term outcomes
CR	Effect of prehabilitation and less invasive bypass on long-term functional and MACE outcomes
Mental health	Prevalence and long-term impact of (underdiagnosed) depression Effect of off-pump and no-touch surgical techniques on perioperative neurological events
All	Importance of patient-centered perspectives in weighing trade-offs between risk and benefit, particularly for riskier interventions

AF indicates atrial fibrillation; CABG, coronary artery bypass grafting; CR, cardiac rehabilitation; DAPT, dual antiplatelet therapy; MI, myocardial infarction; and OAC, oral anticoagulation.

has improved significantly since the 2015 statement from a scope, evidence, applicability, and, perhaps most important, adoption point of view; data suggest a greater uptake worldwide.^{2,4,176–180} It is now well demonstrated

that medication adherence after CABG leads to better long-term survival and outcomes.^{2,5,178,180} During that time, CABG has made a resurgence as the most robust and durable form of myocardial revascularization.¹⁸¹ CABG is also becoming less invasive and more often is combined with interventional and advanced medical therapies,¹⁸² for which the tailoring and optimization of secondary prevention strategies will be particularly germane to successfully enable these new myocardial revascularization approaches and to improve outcomes for future patients with CAD.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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Disclosures

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This table represents the relationships of writing group members that may be perceived as actual or reasonably perceived conflicts of interest as reported on the Disclosure Questionnaire, which all members of the writing group are required to complete and submit. A relationship is considered to be "significant" if (a) the person receives \$5000 or more during any 12-month period, or 5% or more of the person's gross income; or (b) the person owns 5% or more of the voting stock or share of the entity, or owns \$5000 or more of the fair market value of the entity. A relationship is considered to be "modest" if it is less than "significant" under the preceding definition.

*Modest.

†Significant.

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